

CALVIN KWON

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EDUCATION

University of Virginia, School of Engineering and Applied Science

B.S. Computer Engineering

GPA: 3.9

Charlottesville, VA

Expected May 2029

RELEVANT PROJECTS

Gatekeeper - Open-Air Network Vulnerability Detection API

Jan 2025 - Mar 2025

- Team Lead: Managed a team of 6 to design and deploy a full-stack security platform as part of the Microsoft Garage student pilot program.
- Developed a web-based UI and API to automate vulnerability detection in open-air networks, leveraging the GPT-4o API via Node.js to analyze complex network data and identify potential security breaches.

Qiskit - Quantum Random Number Generator Analysis

Sept 2024 - Feb 2025

- Benchmarked Quantum vs. Pseudo-Random Number Generators by implementing a QRNG model in Python and Qiskit.
- Validated higher entropy scores across 100+ trials, proving increased resistance to predictive attacks compared to traditional algorithmic generators.

BoatMeter - Real Time Race Tracking Solutions

Jun 2023 - Dec 2024

- Engineered an end-to-end rowing race tracking system, currently protected by two pending patents, integrating Arduino-based IoT sensors with a real-time FlutterFlow UI.
- Developed Node.js and Firebase backend to manage live telemetry and performance analytics.
- Successfully launched and tested the application during club team sessions on the Occoquan River

WORK EXPERIENCE

Research Intern, Virginia Tech Innovation Campus, Alexandria, VA

Summers 2023 - 2025

- Conducted a study into Lattice-Based Post-Quantum Cryptography under the mentorship of Dr. Jim Egenreider, culminating in a comprehensive capstone presentation.
- Facilitated integrated technology projects in support of K-12 STEM initiatives

HONORS AND AWARDS

- Raytheon (RTX) Invention Convention US Nationals: 1st, 2nd
- Raytheon (RTX) Invention Convention Americas, 2nd
- USRowing U19 Selection Development Program - Selected as one of five U.S. men's coxswains nationwide for elite development training under Olympians and national team coaches.
- Eagle Scout of America

CERTIFICATIONS

CompTIA Security+ (2025), CompTIA A+ (2024), AWS CCP (2025), Microsoft SC-900 (2024), Microsoft AZ-900 (2024), ITS Cloud Computing (2025), ITS Network Security (2025), ITS Cybersecurity (2025), Testout Security Pro (2025), Testout PC Pro (2025).

EXTRACURRICULARS

Head Freshman Coxswain, Virginia Men's Rowing. 20h/week

Owner, KCUTVA, operates an independent mobile haircutting business serving UVA and Nova area, 85+ clients. 8h/week

Stage Director, TedX at UVA. 2h/week

Video Editor, V-Magazine at UVA. 2h/week